

DISASTER RESPONSE · DECISION-SUPPORT PLATFORM

Turning disaster noise into a decision

An evidence-linked operational picture for emergency coordinators — built from fragmented, contradictory, fast-decaying reports, and never hiding what it does not know.

PROBLEM	SOLUTION	IMPACT & VALUE
Information fusion & decision-latency gap	Evidence-linked situation picture + resource matching	Faster, explainable, human-in-the-loop decisions

BUILT BY TEAM **SHUTDOWN** · OPEN INNOVATION TRACK

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Every figure traces to published evidence: CAG performance audits (Karnataka & Kerala), MHA / NDMA situation reports, Kerala SDMA records, and state coordination orders, 2024–2026.

I · THE PROBLEM

The bottleneck is not missing infrastructure

It is [information velocity, fragmentation, uncertainty and change](#).

During floods, landslides and other disasters, India's emergency authorities already receive information from weather agencies, hydrological sensors, CCTV, police, rescue teams, hospitals, utilities, local officials, citizens, NGOs and volunteers.

Within minutes, one operations centre can receive a flood warning, a blocked-road report, a trapped-resident report, a damaged-bridge report, a hospital evacuation request, a power-outage report — and a contradicting update. Someone has to manually reconcile all of it into one current picture.

REFINED PROBLEM STATEMENT

Emergency coordinators must manually reconcile reports arriving through different channels, identify duplicate or contradictory information, verify whether routes and resources are still available, and repeatedly rebuild the situation picture as conditions change — creating avoidable decision latency and raising the risk that resources are delayed, duplicated, or routed through inaccessible paths.

II · THE PROOF — CASE 01

Karnataka built the sensing layer. The operational loop still failed.

49

of 100 sensors

Flood water-level sensors on Bengaluru storm-water drains were **non-functional** as of December 2023 — so the data the flood model depended on never arrived.

BEYOND BENGALURU

69 of 184

Rain gauges, weather stations and water-level sensors across Mangaluru, Belagavi and Hubballi-Dharwad were also non-functional.

PUBLIC MONEY AT STAKE

₹2.45 crore

Flood-model and sensor investment rendered largely unfruitful because the expected data never reached the responsible agency.

Source: Comptroller & Auditor General of India, Report No. 08 of 2025 on Disaster Management in Karnataka.

II · THE PROOF — CASE 02

Kerala 2018: agencies existed. The information chain broke.

COMMUNICATIONS

Total failure

Idukki and Chalakudy taluk officials reported complete communication failure in their areas; infrastructure failed at some dam sites and government offices during and after the floods.

IDUKKI RESERVOIR RULE CURVE

1983

The operating rule curve dated from 1983 and had never been reviewed — decisions ran on a 35-year-old assumption set.

IDAMALAYAR RESERVOIR

None at all

No operating rule curve existed for the reservoir, leaving release decisions without a documented reference.

WHAT BOTH AUDITS CONVERGE ON

The loop broke at data handling, not at hardware.

Sensors, telemetry and a planned urban flood-forecasting model existed for Bengaluru — but sensor locations and outputs were often unavailable to the responsible agency, and the model was never effectively implemented. That sets our differentiator: sensor-health monitoring, missing-data alerts and confidence indicators — not another map.

Sources: CAG Performance Audit of Kerala Flood Preparedness & Response, Report 6 of 2021; CAG Report No. 08 of 2025, Preface & Executive Summary.

III · ROOT CAUSES & USERS

Ten root causes, one decision-support gap

It sits between raw reports and coordinated field action — the reasoning workload, not the data collection.

- | | |
|---|---|
| 01 Multi-source information arrives through different channels and formats. | 06 Resource availability is dynamic, not static. |
| 02 Information quality and freshness vary significantly. | 07 Geographic accessibility decides whether help can actually arrive. |
| 03 Multiple reports can describe the same incident — duplication. | 08 Communication networks become unreliable exactly when needed most. |
| 04 Reports conflict — different observers, different moments in time. | 09 NGOs and volunteers add capability but need coordination to be useful. |
| 05 Ground conditions change after a decision is already made. | 10 Operators must decide fast while processing huge information volume. |

PRIMARY USER

The emergency coordinator or operations officer inside a state, district or local Emergency Operations Centre — SEOC / DEOC.

SECONDARY USERS

- Incident commanders and field coordinators
- Search-and-rescue teams — NDRF / SDRF
- Police and fire services
- Medical and hospital coordinators
- Public-works and utility coordinators
- NGO and community coordination personnel

A decision-support product for people who already hold authority and operational responsibility — not a public chatbot instructing citizens or field teams directly. A conventional dashboard can display data; a coordinator must still work out whether two reports are the same incident, which is newer, whether a road is reachable, and whether an earlier decision now needs revisiting.

IV · THE SOLUTION

An information-fusion and decision-support layer – not a replacement for command

OBJECTIVE

Turn fragmented, changing and uncertain disaster information into an evidence-linked operational picture that helps emergency coordinators prioritize incidents and resources faster.

01

Input

Heterogeneous reports enter from all six source layers.

02

Normalize

Converted into one common incident schema.

03

Correlate

Duplicate and related reports link into a single incident.

04

Validate

Conflicts and stale data are detected explicitly, never silently resolved.

05

Assess

Accessibility, resource feasibility and confidence are scored.

06

Prioritize

An explainable ranking of what needs attention now.

07

Coordinate

Recommendation plus its evidence shown to a human coordinator.

08

Update

New reports revise the picture and flag affected decisions.

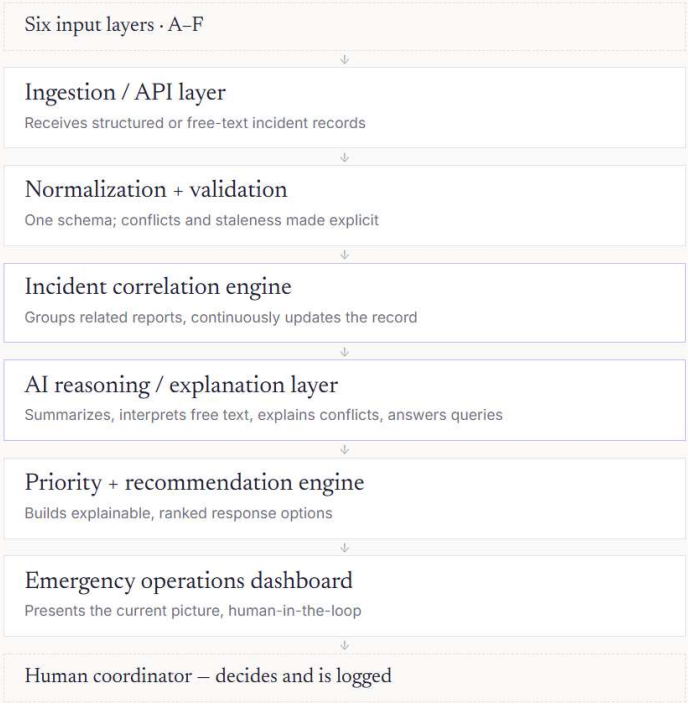
IV · INPUT LAYERS

Six input layers, modelled on a real Gujarat response environment

LAYER	EXAMPLES	PURPOSE
A · Warning & environmental	Weather and rainfall alerts, river levels, GIS and satellite data	Establish hazard context and likely-affected areas
B · Emergency & field reports	Police, fire and rescue, SDRF / NDRF, taluka reports	Direct ground-truth observations
C · Medical	Hospital capacity, casualties, evacuation requests, ambulances	Estimate humanitarian and medical urgency
D · Infrastructure & utilities	Road and bridge status, power, water, telecom, public works	Accessibility and secondary-risk assessment
E · Community & NGO	Citizen reports, volunteers, NGO requests and offers, shelters	Local visibility beyond official channels
F · Resources	Rescue teams, boats, ambulances, relief material, shelters	Connect incidents to real response capability

The prototype uses structured or simulated data representing these sources wherever live government feeds are unavailable. No claim of access to protected systems is made.

Deterministic logic where certainty exists. AI where language and judgement are required.



DETERMINISTIC LOGIC HANDLES

- Timestamps and unique incident IDs
- Source metadata tracking
- Resource availability state
- Basic geographic calculations
- Status-transition rules
- Validation rules

AI ASSISTS WITH

- Extracting structured facts from free-text reports
- Semantic incident matching and correlation
- Summarizing large evidence sets
- Explaining contradictory observations
- Answering natural-language operator questions
- Generating human-readable situation briefs

PERSISTENT LAYERS THE CORRELATION ENGINE READS AND WRITES

Evidence store Every original report and timestamp preserved — never overwritten by an AI summary	Geospatial layer Locations, routes, affected regions, live resource positions	Rules / analytics Deterministic status, timers and thresholds
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Development workflow: built, reviewed, tested and documented using IBM Bob — Ask, Agent and Plan modes, code review, terminal and subagent workflows — as the core engineering layer.

VI · COORDINATOR EXPERIENCE

Five screens, each answering one question a coordinator actually asks

Evidence first · uncertainty visible · action-oriented · human-in-the-loop · low cognitive load.

SCREEN	ANSWERS	SHOWS
1. Situation overview	<i>“What is happening right now?”</i>	Incident map, critical and unverified incidents, resource and accessibility status
2. Incident workspace	<i>“Why do we believe this?”</i>	Supporting versus conflicting evidence, confidence, access route, status
3. Priority & action	<i>“What needs attention now, and why?”</i>	Ranked incidents with an explicit, readable reason for each rank
4. Resource view	<i>“What can realistically respond?”</i>	Available resources, assignments, location, capability
5. Change alert	<i>“What just changed?”</i>	New evidence, plus which existing response plans it may invalidate

VI · REHEARSED SCENARIO

A Gujarat district flood, end to end

01 · Warning

Elevated flood risk triggers the initial operational context.

02 · Reports arrive

Police report a blocked road; a citizen reports a stranded village; a hospital needs 12 patients relocated; a field team calls a bridge unsafe; an NGO requests food; a utility reports a power outage.

03 · Correlation

The system links all six reports to one geographic incident cluster.

04 · Conflict

A new citizen report claims the road is open. The system marks accessibility **uncertain** — it does not overwrite the field report.

05 · Resource matching

An available boat team and an ambulance are linked to the stranded village.

06 · Prioritization

Hospital evacuation is ranked highest, with the reasons shown explicitly.

07 · Operator action

The coordinator accepts, edits or rejects the recommendation; the decision is logged.

08 · Situation changes

A new report says the bridge is deteriorating further; the system flags every response plan that depended on it.

One rehearsed scenario, designed to demonstrate the problem, the solution, the UX and the impact in a single coherent narrative.

VII · IMPACT & VALUE

Value comes from cutting decision latency — **not from replacing responders**

EXISTING FLOW

- ↓ Many incoming reports
- ↓ Human collection
- ↓ Manual comparison
- ↓ Manual interpretation
- ↓ Manual prioritization
- ↓ Resource coordination
- ↓ Field action

PROPOSED FLOW

- ↓ Many incoming reports
- ↓ AI-assisted ingestion + normalization
- ↓ Incident correlation
- ↓ Conflict + freshness detection
- ↓ Evidence-linked situation picture
- ↓ Explainable prioritization
- ↓ Human decision → field action
- ↓ Continuous update on new reports

WHERE VALUE IS CREATED

- Identifies **duplicate reports** before they waste responder attention
- Exposes **contradictions** instead of silently trusting the latest message
- Surfaces **stale information** so it is never mistaken for current truth
- Connects incidents to **feasible, reachable resources**
- Flags **inaccessible routes** before a team is dispatched
- Keeps operational status current as the disaster evolves
- Shows the **evidence behind every recommendation**, not a black-box score
- Cuts time spent manually rebuilding the situation picture from scratch

VII · WHAT WE WILL MEASURE

Reported as prototype test results – not as proven deployment outcomes

INFORMATION PROCESSING

- Time to consolidate a defined report set
- Duplicates detected · conflicts surfaced · stale reports identified

DECISION SUPPORT

- Time from incident arrival to prioritized view
- Share of high-priority incidents correctly surfaced · resource-match accuracy
- Proportion of decisions made with visible evidence

CHANGE MANAGEMENT

- Time for the dashboard to reflect a new report
- Affected recommendations correctly re-flagged after a major update

WHY CHANGE-REFLECTION LATENCY IS A METRIC

Himachal Pradesh, 2026: the blocked-road count moved inside a single day.



Blocked roads reported in Himachal Pradesh, 2026. Source: Times of India, via the project evidence set. A situation picture that refreshes daily is already 12 roads out of date.

VIII · EVIDENCE-TO-FEATURE MAP

Every feature traces back to a documented failure — not a hypothetical

DOCUMENTED FINDING	REQUIRED FEATURE
49/100 Bengaluru water-level sensors non-functional (CAG)	Sensor-health monitoring & missing-data alerts
5 Bengaluru street sensors had no available data at all (CAG)	Coverage-gap & last-update visibility
Communications failed in parts of Idukki & Chalakudy, Kerala 2018 (CAG)	Multi-channel ingestion, offline queue, comms-status tracking
Idukki reservoir rule curve unreviewed since 1983; none existed for Idamalayar (CAG)	Versioned rules, live readings, decision-change alerts
Wayanad 2024: bridge washed away, access severed for ~2,000 people (MHA / ReliefWeb)	Route accessibility & resource-feasibility analysis
~65,000 rescued by 4,537 fishermen with 669 boats, Kerala 2018 (Kerala SDMA)	Verified community / volunteer resource registry
Assam 2026: relief groups converged on some areas while others stayed underserved (Assam Tribune)	Demand-supply matching & underserved-area detection
Himachal Pradesh 2026: blocked-road count changed from 61 to 49 within one day (Times of India)	Timestamped, live accessibility updates
Vadodara 2024: 8,361+ evacuated in a multi-agency response (Indian Express)	Explainable prioritization & resource-allocation view

VIII · HONEST POSITIONING

A reliability-aware information-fusion and resource-orchestration layer for emergency operations – not a flood map, not a chatbot, not an autonomous dispatcher, not a replacement for control rooms.

WE CLAIM

- An AI-assisted information-fusion and decision-support layer
- Transforms heterogeneous reports into evidence-linked intelligence
- Demonstrably faster consolidation, conflict detection and resource matching in a controlled scenario

WE DO NOT CLAIM

- Replacing emergency authorities or NDRF / SDRF command
- Autonomous dispatch of rescue assets
- Access to confidential government feeds we do not actually have
- Proven casualty or response-time reduction without real deployment data

Decision requested from reviewers: approve the rehearsed Gujarat district-flood scenario as the prototype's demonstration scope, and the evidence-to-feature map on the previous page as the build backlog. Every item on that backlog is traceable to a published audit or situation report.